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Multiplexed single-cell 3D spatial gene expression analysis in plant tissue using PHYTOMap

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Retrieving the complex responses of individual cells in the native three-dimensional tissue context is crucial for a complete understanding of tissue functions. Here, we present PHYTOMap (plant hybridization-based targeted observation of gene expression map), a multiplexed fluorescence in situ hybridization method that enables single-cell and spatial analysis of gene expression in whole-mount plant tissue in a transgene-free manner and at low cost. We applied PHYTOMap to simultaneously analyse 28 cell-type marker genes in *Arabidopsis* roots and successfully identified major cell types, demonstrating that our method can substantially accelerate the spatial mapping of marker genes defined in single-cell RNA-sequencing datasets in complex plant tissue.

Understanding how individual cells respond and interact with each other in the face of changing environments is the cornerstone of understanding tissue function. Single-cell transcriptomics technologies have been widely adopted in plant research, enabling the classification of cells into populations that share molecular features for the in-depth analysis of cell types and states[1–3]. Increasing throughput and sensitivity in single-cell transcriptomics technologies will offer tremendous granularity at which cells can be classified, but will also create new challenges in dealing with cell populations that our current histological and physiological understanding of plant cells cannot account for. To understand the identity and function of molecularly defined cell populations, it is critical to analyse their spatial localization in the tissue.

In plant research, the most common tool for spatially mapping cell population marker genes identified in single-cell transcriptome analysis has been transgenic reporter lines that express fluorescent proteins under the predicted promoter region of the genes. In most cases, each transgenic line visualizes the expression of only one gene. This approach has several limitations when analysing cells in complex tissue: (1) a cell type/state is not always defined by the expression of a single gene, but by the combination of many genes; (2) spatial mapping of a single gene or a few genes has difficulties in analysing multiple cell types/states simultaneously, which is critical for understanding interactions between cell types/states; (3) generation of transgenic plants is time-consuming; and (4) heterologous expression of fluorescent proteins does not necessarily reflect the true expression of the gene because the reporter cassettes lack the native genomic context (for example, enhancer–promoter interactions). In situ hybridization, another popular approach in spatial gene expression analysis in plants[4], can overcome a few of the above limitations but suffers from low multiplexing capacity. Therefore, spatial gene expression analysis needs to be done with a large number of genes at single-cell resolution for a more complete understanding of the function of cell types/states and their interactions with other cells and the environment.

Spatial transcriptomics technologies hold great promise in addressing these problems by simultaneously revealing the molecular details and spatial location of cells in complex tissues. Methods using spatially barcoded arrays or imaging-based, highly multiplexed single-molecule fluorescence in situ hybridization allow researchers

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